1. Aspect 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Analysis**: The problem explicitly states that "the posters below show how they pay their sellers." Students must extract the payment rules from the poster image (e.g., commission per newspaper, fixed weekly salary plus bonus, etc.). This is the first and most critical step in modelling. Without correctly decoding the visual information, a mathematical model cannot be built.

2. Aspect 2: Identify the key variables in a model

- **Analysis**: The core variables in this problem are "weekly sales quantity" (given as 350 copies) and "weekly earnings" (to be found). Additionally, the payment rules in the poster may introduce other variables such as a fixed base salary, unit commission rate, or tiered bonuses. Students must distinguish between input and output variables.

3. Aspect 6: Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Analysis**: This problem transforms a real-world situation (newspaper seller compensation) into a mathematical problem. Students need to recognise that earnings are a function of sales quantity (usually linear or piecewise linear) and distinguish between fixed and variable components. This demonstrates the ability to abstract mathematical structure from a real context.

4. Aspect 5: Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Analysis**: The payment scheme typically takes the form "Total pay = Fixed part + Unit commission × Sales quantity" or a similar linear relationship. Students must identify this linear structure and apply it. If the poster includes tiered commissions (e.g., higher rate after 200 copies), they also need to recognise a piecewise pattern.

5. Aspect 8: Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **Analysis**: This problem is of the type "given unit rate and total quantity, find total earnings", corresponding to multiplication or linear function applications at elementary or middle school level. Students need to connect it to prior experience (e.g., "unit price × quantity = total cost") to select the correct operation.

6. Aspect 9: Translate a problem into a standard mathematical representation or algorithm

- **Analysis**: Once the payment rules are understood, students must express them as a mathematical formula (e.g., `Earnings = 50 + 0.2 × 350`) and perform arithmetic operations. This is a key step from textual/visual description to symbolic computation.

7. Aspect 3: Select a representation appropriate to the problem context

- **Analysis**: Students need to decide whether to use a table, formula, or graph to represent the relationship between earnings and sales. For a straightforward calculation, an algebraic expression is usually sufficient; however, if the poster rules are complex, a piecewise function might be needed. Choosing the right representation helps ensure accurate calculation.

8. Aspect 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **Analysis**: The problem provides an average sales figure and implicitly assumes stable weekly sales (ignoring fluctuations). Students may also ignore real-world details such as taxes, day-to-day variations, or returns, focusing only on the payment rules explicitly given in the poster. This simplification is a necessary part of modelling.

9. Aspect 10: Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- **Analysis**: Students may introduce symbols (e.g., let `x` be sales quantity, `y` be earnings) and write an equation `y = a + b·x`. Although the problem can be solved with direct arithmetic, more complex rules may require symbolic representation, which is an application of mathematical tools.

10. Aspect 1: Select a mathematical description or a representation that describes a problem

- **Analysis**: Students must choose from several possible mathematical forms (e.g., proportion, linear function, piecewise function) the one that best matches the poster's rules. For example, if the poster shows "0.5 per copy", they select a direct proportion model; if it shows "base salary + commission", they select a linear model.

1. 方面4：阅读、解码并理解陈述、问题、任务、对象或图像，以创建情境模型

- **分析**：题目中明确提到“下面的海报显示了他们如何支付销售员”，学生必须从海报图像中提取支付规则（例如，每份报纸的佣金、固定周薪加提成等）。这是解题的第一步，也是最关键的建模活动。没有正确解码图像信息，就无法建立数学模型。

2. 方面2：识别模型中的关键变量

- **分析**：题目涉及的核心变量包括“每周销售数量”（已知为350份）和“每周收入”（待求）。此外，海报中的支付规则可能引入其他变量，如固定底薪、单位提成率、阶梯奖励等。学生需要明确哪些变量是输入、哪些是输出。

3. 方面6：识别并描述现实问题情境中的数学方面，包括确定重要变量

- **分析**：该问题将报纸销售员的薪酬计算转化为数学问题。学生需识别出薪酬与销售量之间存在函数关系（通常是线性或分段线性），并区分出固定部分与变动部分。这体现了从现实情境中抽象数学结构的能力。

4. 方面5：识别问题或情境中的数学结构（包括规律、关系和模式）

- **分析**：支付方式通常呈现为“总报酬 = 固定部分 + 单位提成 × 销售量”或类似形式。学生需要发现这种线性关系，并运用它进行计算。如果海报包含阶梯提成（如超过200份后提成增加），则还需识别分段模式。

5. 方面8：识别问题中与已知问题或数学概念、事实、程序相对应的方面

- **分析**：该问题属于“给定单位量和总量，求总报酬”的类型，对应小学或初中阶段的乘法或线性函数应用。学生需将其与已有经验（如“单价×数量=总价”）联系起来，从而选择正确的运算。

6. 方面9：将问题转化为标准的数学表示或算法

- **分析**：一旦理解支付规则，学生需将其表达为数学公式（如 `Earnings = 50 + 0.2 × 350`），并执行算术运算。这是从文字/图像描述到符号计算的关键转化步骤。

7. 方面3：选择适合问题情境的表示方式

- **分析**：学生需要决定用表格、公式还是图形来表示薪酬与销售量的关系。对于简单计算，通常直接使用代数表达式；但若海报规则复杂，可能需要分段函数。选择恰当的表示有助于准确计算。

8. 方面7：简化或分解情境或问题，以使其适合数学分析

- **分析**：题目已给出平均销售量，并隐含假设每周销售稳定，忽略波动性。学生也可能忽略税收、工作日差异等现实细节，只关注海报明确给出的支付条款。这种简化是建模的必要部分。

9. 方面10：使用数学工具（变量、符号、图表）描述问题中的数学结构和/或关系

- **分析**：学生需引入符号（如设 `x` 为销售量，`y` 为收入），并写出方程 `y = a + b·x`。虽然本题直接计算即可，但更复杂的规则可能需要符号化表达，这属于数学工具的运用。

10. 方面1：选择描述问题的数学描述或表示

- **分析**：学生需要从多种可能的数学形式（如比例、线性函数、分段函数）中选择最贴合海报规则的那一种。例如，若海报显示“每份0.5元”，则选择正比例模型；若显示“底薪+提成”，则选择线性模型。